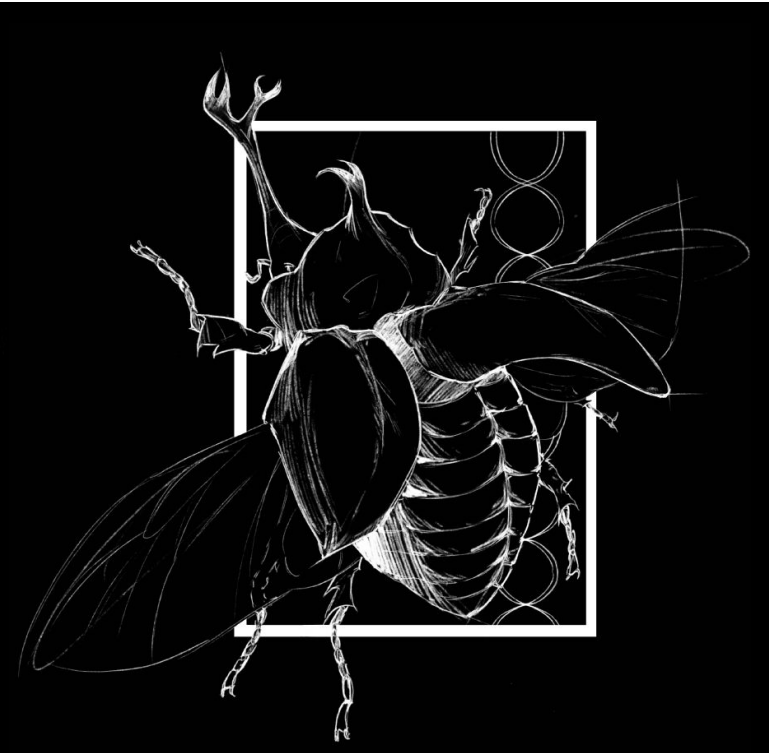




Background

Hymenoptera is large clade that is composed of bees, ants, sawflies, and wasps. Feeding habits among these groups vary widely, ranging from scavengers to predators. Some in this clade, particularly wasps, exhibit parasitoid behavior, where the larvae develop inside of a host species which eventually culminates in the host's death. Many times, these parasitoids are specialized for a particular host species, leading to more fragmented populations and a smaller effective population size (N_e). With smaller effective populations, genetic drift takes precedence over natural selection and deleterious mutations will have a reduced chance of being eliminated. **This is the main question of the Drift Barrier Hypothesis, which proposes that karyotype diversity is driven by genetic drift rather than natural selection.**

Since parasitoid Hymenoptera have a smaller effective population size, it can then be hypothesized that they will also have a higher karyotype diversity. Using data on the parasitoidism of species in the Hymenoptera, as well as haploid chromosome number across the clade, we can compare these variables to determine if species that are parasitoids have a higher karyotype diversity than non-parasitoids. By studying the parasitoid data against genome size, we can determine how effective population size affects how chromosomal changes accumulate over time in the Hymenoptera.

Study Design & AI

In this study, it was investigated of how the state of being parasitoid or non-parasitoid has affected the individual rate of haploid chromosome number evolution across the Hymenoptera. The hypothesis formed was that parasitoid species in the Hymenoptera will have a higher rate of chromosome number evolution compared to that of the non-parasitoid species. The first batch of data that was collected was the haploid chromosome number for the species in Hymenoptera, and this data was provided by Dr. Blackmon. I also found data of whether the studied species in Hymenoptera were parasitoid or non-parasitoid from the Nature Communications Journal. In the data, the state of being parasitoid or non-parasitoid was separated into the states of 0 (non-parasitoid) or 1 (parasitoid). In this way, being parasitoid or non-parasitoid was able to be analyzed as a binary trait that was then compared to chromosome number, allowing for the ChromPlus model to be used.

LLMs were used several times throughout this project. Large language models such as chat.tamu.ai were utilized to gather information about the chosen clade, as well as compiling sources for the requested data. All of the information provided by the LLM was checked and re-verified by a human source. The LLM was also used to generate code that created a csv file of the species list which would later be used to compile the data collected. After the data had been compiled, the LLM was further used to generate a code that produced the figures above.

Model Fitting

In this project, ChromPlus was used. This model assumes that, rather than there being only one rate of chromosomal evolution across species, there are instead two. These two rates pertain to a defined binary trait where there are separate rates of fusion, fission, and polyploidy for each state. This allows for the clear distinction between lineages that have evolved rapidly versus lineages that have evolved slower, and enables us to test hypotheses concerning how the presence or absence of a trait may affect karyotype diversity. In this case, ChromPlus was used to directly compare the chromosome number evolution rate between parasitoid and non-parasitoid lineages in Hymenoptera.

Along with ChromPlus, Bayesian Inference (MCMC) was also used. The Bayesian Inference model estimated and tested a range of possible parameters. It was used specifically in this study since it assumes that there are two distinct groups of species within the clade following a binary trait. It runs Markov Chain Monte Carlo (MCMC) model to test thousands of possible parameter values that could explain what is observed in the tree before building a distribution of the most plausible answers. This model was used instead of the Maximum Likelihood model since it takes into account the uncertainty and avoids making assumptions of the shape of the data.

Results

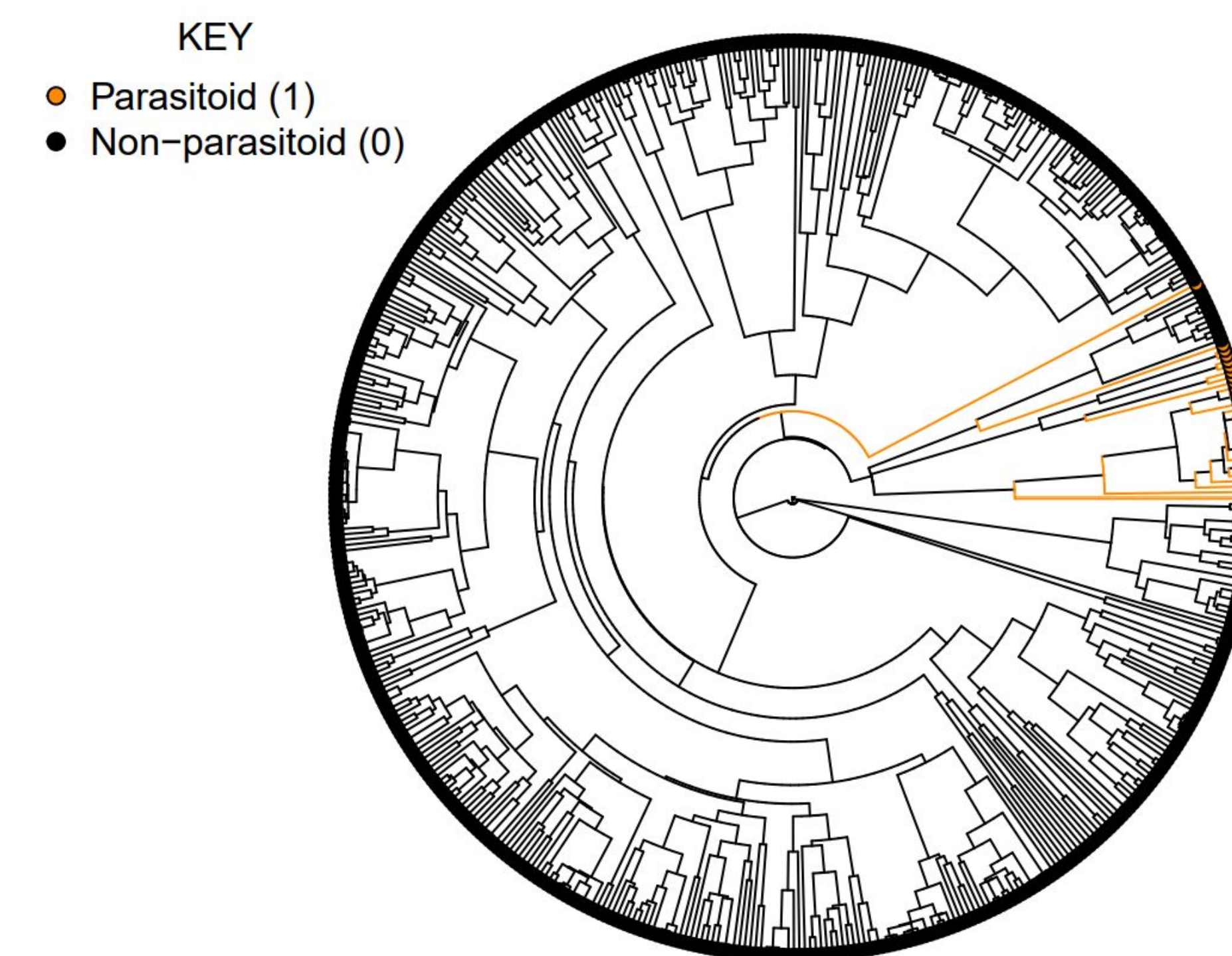


Figure 1. A phylogenetic tree in a circular format that has colored circles at the end of the tree tips. The black circles represent non-parasitoid species while the orange circles represent the parasitoid species.

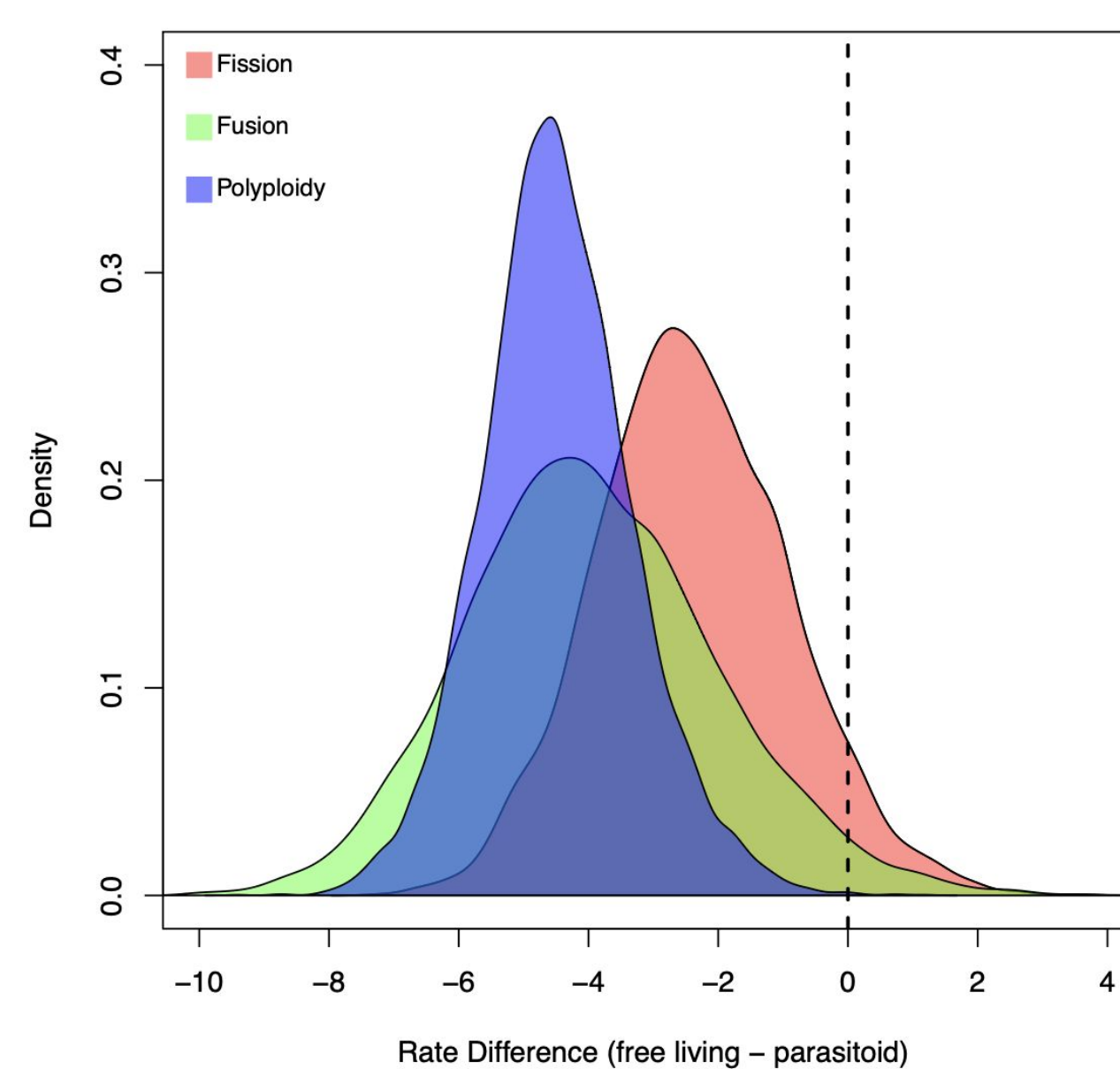


Figure 4. A posterior plot with rate differences on the x-axis and density on the y-axis. Each distributions represent the distribution of rate differences (free living minus parasitoid). The fact that each distribution is almost entirely below zero indicates that parasitoid lineages have increased rates of fusion fission and polyploidy.

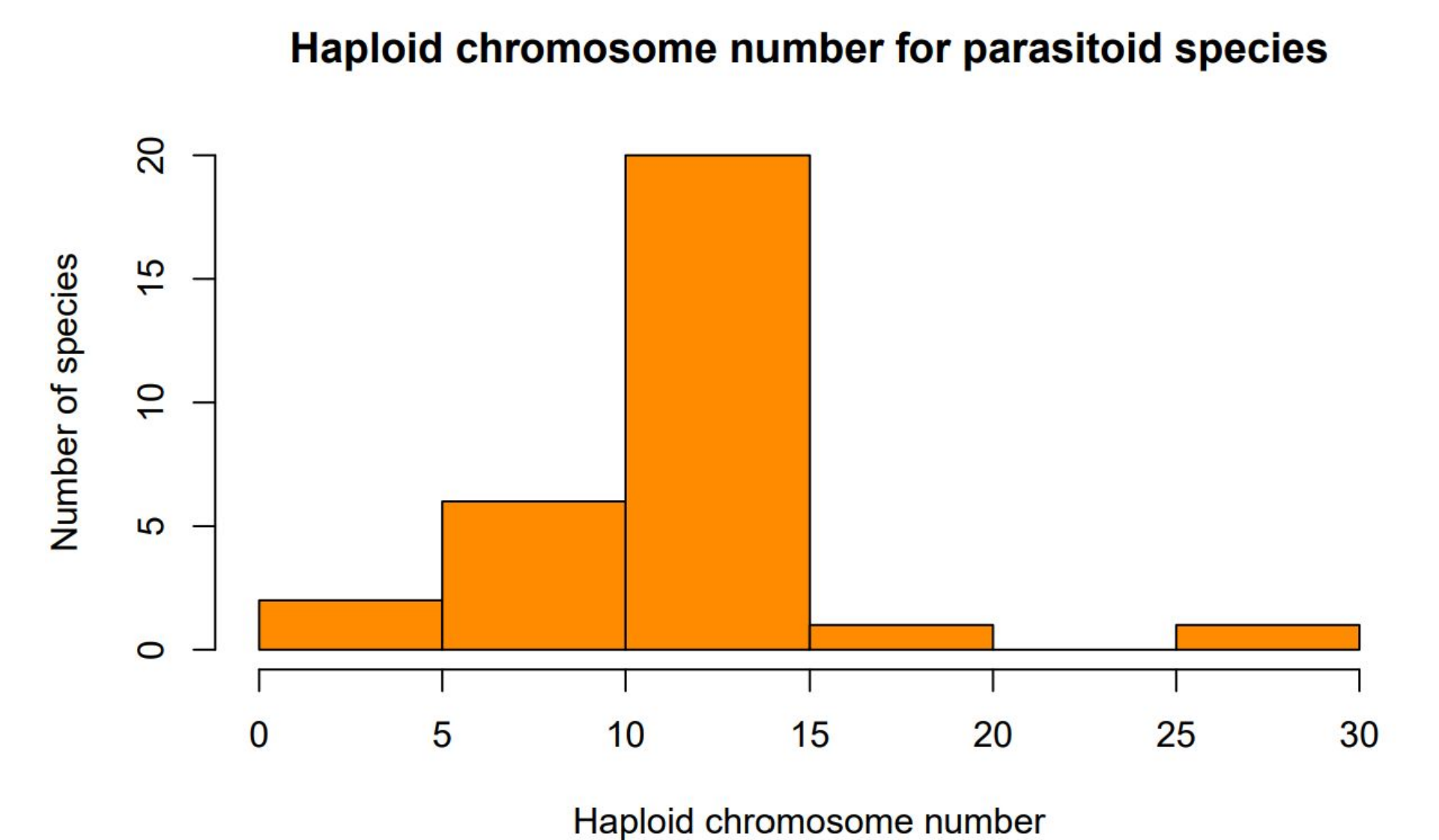


Figure 2. A histogram depicting the haploid chromosome number distribution among the parasitoid species of Hymenoptera.

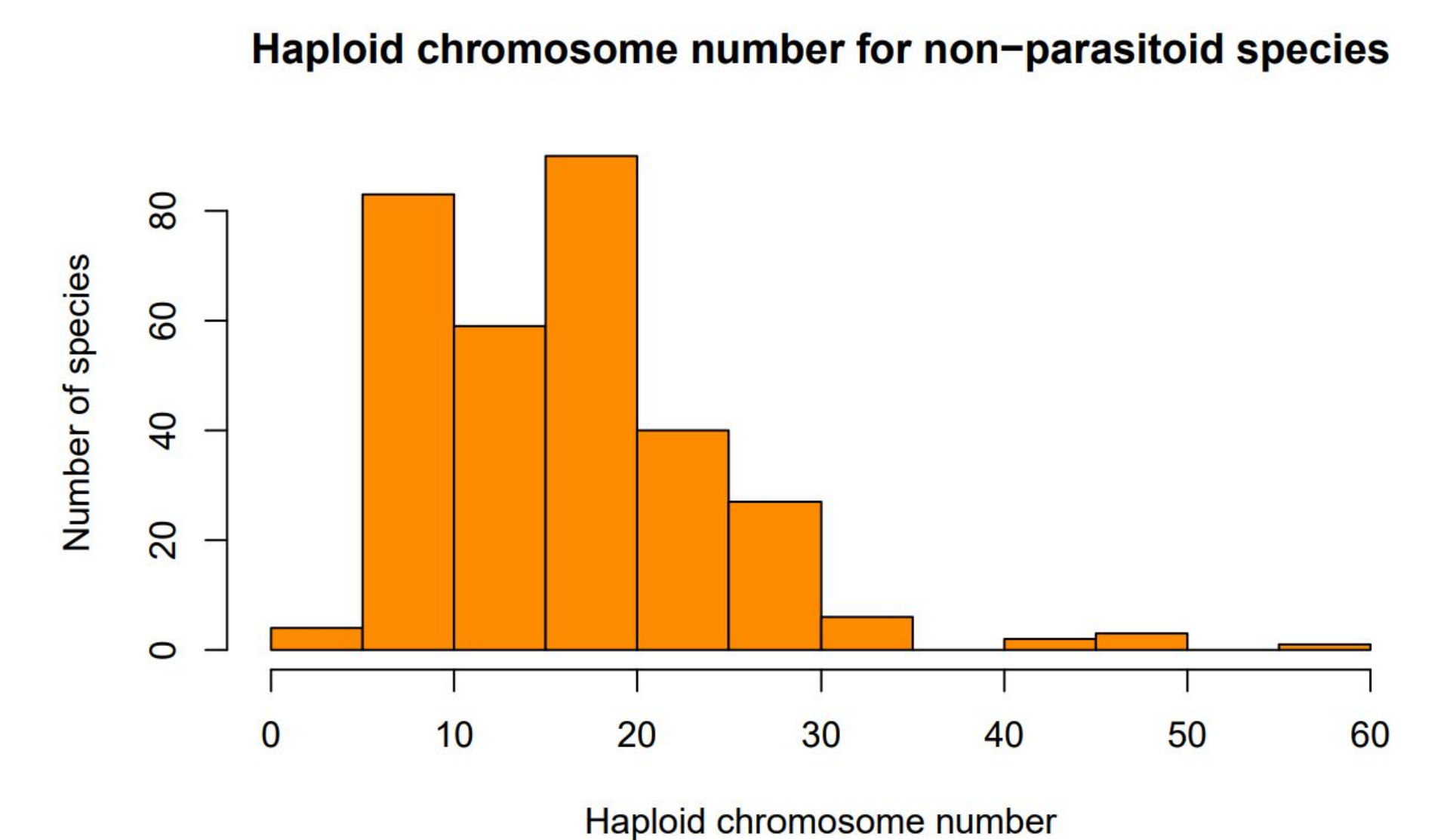


Figure 3. A histogram depicting the haploid chromosome number distribution among the non-parasitoid species of Hymenoptera.

Discussion

My hypothesis proposed that parasitoid species in the Hymenoptera would exhibit a higher rate of chromosome number evolution compared to non-parasitoid species. According to the Drift Barrier Hypothesis, this would be plausible, as parasitoids are likely to have a smaller effective population size than non-parasitoids. The data collected supports my hypothesis, though some parameters are less strongly supported than others.

There is strong evidence suggesting that parasitoid species have a higher rate of polyploidy when compared to non-parasitoids. The calculated 95% confidence interval from ChromPlus for non-parasitoids is 0.59-4.02, while the parasitoid's interval is 4.95-8.14. Since there is no overlap at all, this provides strong support that parasitoids exhibit a higher rate of polyploidy than non-parasitoids. This can also be seen in Figure 4, where the polyploidy rate difference distribution is entirely negative. For the remaining parameters, such as the rate of fission and fusion, the results only provide moderate support. The 95% confidence interval for the rate of non-parasitoid fission and parasitoid fission do overlap, although the parasitoid fission distribution has a consistently higher parameter value. This can be seen with how the mean for the parasitoid fission distribution is 5.18 while the mean for the non-parasitoid distribution is 2.79. This is a similar case with the fusion distributions, although the overlap is less severe than that of the fission distributions. The parameter values for the parasitoid fusion distribution were also consistently higher than the non-parasitoid distribution, with their means being 7.76 and 3.80 respectively. In Figure 4, both rate difference distributions for fission and fusion have an area where they are positive, but a majority of their distributions are negative. This further indicates that there is support for the idea that parasitoid lineages have a higher rate of chromosome number evolution; however, it is only moderate.

It should also be noted that the rate of transitioning from being non-parasitoid to parasitoid is consistently much higher than the rate of transition from a parasitoid to non-parasitoid state. Their respective means are 0.73 and 0.031. The very small value for tran21 suggests that once a species has transitioned from being non-parasitoid to parasitoid, it is very unlikely that they would transition back. This provides stronger support for the idea that the parasitoid state can significantly shape chromosome number evolution.

However, despite the support found in the results, there are several possible limitations that were found. For one, there were multiple species in the Hymenoptera where data for their chromosome number, state of being parasitoid/non-parasitoid, or both were unable to be found. In particular, many of these species were clustered together in the same family, so the results may be biased towards certain families in the clade Hymenoptera while ignoring others. Another limitation is the unbalanced number of species in each state. Compared to the amount of non-parasitoid species, the number of parasitoid species is very small. This could skew the calculations made for the rates of chromosomal number evolution in parasitoid species as the sample size is much smaller.

In the future, studies investigating this question should aim to fill in the missing data to ensure results that encompass the entirety of the Hymenoptera. Furthermore, they should also take into account the many other evolutionary events that could occur that affect chromosome number, rather than focusing solely on fission, fusion, and polyploidy. Finally, they should try to put more detail into the parasitoid state. Instead of simply categorizing species into either being parasitoid and non-parasitoid, they could also measure the amount of host species for the parasitoids to see if an increasingly restricted number of possible hosts also affects karyotype diversity.

Conclusions & Acknowledgements

In conclusion, my hypothesis is supported as the values for the rates of chromosomal number evolution were consistently higher in parasitoid species than in non-parasitoid species. In particular, there is strong support for the idea that the polyploidy rates in parasitoids are higher than that of non-parasitoids. Thank you to Dr. Blackmon and my classmates for their valuable assistance and insight throughout this project!